

Identifying post-suburbanization: The case of the Jakarta metropolitan area (JMA)

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ABSTRACT

The post-suburban notion suggests the transformation of suburbs due to infrastructure expansion and connectivity between primary and smaller cities, which leads to polycentricity. This paper examines post-suburbanisation by examining commuting patterns, origin-destination, and commuters' socio-economic characteristics in the Jakarta Metropolitan Area (JMA). This study uses cross-sectional commuting survey data in 2014 and 2019 to investigate the post-suburbanisation process by three indicators; (i) spatial dependence of job and school activities to Jakarta is limited, (ii) maturity of sub-urban is higher in outer districts as it is more attractive for middle-upper class and older migrants and (iii) segregation as a higher share of commuters from adjacent districts. This paper contributes to the post-suburbanisation literature by highlighting the importance of commuting data and the dynamic of urban socio-spatial in a developing country that calls for multilevel institutional arrangements to manage the metropolitan.

1. Introduction

The shift of development from large cities to suburbs in the last few years marks the process of post-urbanisation globally. The post-suburban literature has been growing in recent years following the transformation of suburbs and suburban politics with the emergence of numerous outer and edge cities (Firman & Fahmi, 2017; Hudalah & Firman, 2012; Phelps, 2018; Phelps, 2018, 2018). This rise has been due to infrastructure expansion and connectivity between major and smaller cities, as found in the United States with interstate highways and state parkways. Post-suburbanisation is seen as the maturing of suburban territory and sprawl of the settlement, referring to housing that has been ageing, losing its attractiveness to middle- and upper-income persons. As such, selective revival, or gentrification of neighbourhoods, will occur. Thus, planning and design are critical in urban physical adaptation into maturing places and communities.

The case study, Jakarta Metropolitan Area (JMA), has persistently grown economically and spatially for more than four decades. Studies in JMA have shown that the development of the industrial estate, urbanisation, and the emergence of new town residential contributed to the metropolitan expansion and independency of peripherals to core cities (Firman, 2012; Firman & Fahmi, 2017). As such, the large-scale development in the JMA has led to complexity in both actor networks and governance that led to urban issues.

With the growing spatial size and socio-economic activities, there is a lack of study on the post-suburbanisation process by looking at commuting and its spatial implications. There is limited research examining commuters' socio-economic activities and spatial mobilities to understand the post-suburbanisation process. Presently, studies in post-suburbanisation focus on non-spatial aspects, such as the relationship between urban form and spatial structure (Borsdorf, 2004; Helbich & Leitner, 2010; Sadewo, Syabri, Antipova, Pradono, & Hudalah, 2021), political and institutional matters (Phelps, Wood, & Valler, 2010), ecology (Alberti & Marzluff, 2004; Sadewo, Syabri, & Pradono, 2018) and the socio-economic development (Aritenang, 2021, 2022; Indraprahasta & Derudder, 2019; Musil, 2007).

This paper aims to examine the process of post-suburbanisation by looking at spatial independence, maturity, and segregation patterns. Whilst previous studies on post-suburbanisation in JMA uses industrial data, such as labour and spatial distribution, and new large-scale residential development, this study advances the literature by utilising commuting data looking at population mobility between core and peripheral districts within the metropolitan. The JMA Commuting Surveys in 2014 and 2019 allow to identify post-suburbanisation process through commuters' socio-economic characteristics, mobility, and spatial implications in the JMA between the study period.

The structure of the paper is as follows; Section 2 discusses the literature on post-suburbanisation and the growth of JMA. In section 3,

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we present the research design with data and methodology employed, and section 4 provides results from descriptive and spatial analysis. We present the research conclusion and policy implications in section 5.

2. Post-suburbanisation in a global context

The phenomenon of the expansion of the city has pushed the suburbs to become more attractive while the city centre is starting to lose its influence, this is known by several other terms such as 'edge city', 'edgeless city', and 'technoburb' (Fishman, 2002; Lang & Knox, 2009; Phelps, 2012). However, recent literature sees these various terms as part of post-suburban, describing the emergence of new, predominantly employment-based settlements that are distinguished from the established central cities nearby (Wu & Phelps, 2008).

The advancement in technology has led to the post-suburbia shift with the development of automobile technology that allows people to travel across the metropolitan daily, telecommunication and computer technology that may replace the necessity of face-to-face contact (Hudalah & Firman, 2012). The result was the transformation of peripheral areas to be attractive with urban functions such as shopping and high-tech industrial centres, which previously were exclusive to the metropolitan cores (Muller, 1982).

Furthermore, external forces of globalisation of neoliberal ideology shaped by international actions and the global market have reinforced suburbia's transformation. The globalisation notion is an interregional and international competition with the emergence of flexible specialisation where global manufacture relocated in countries that provided the lowest cost of production, both capital and labour. The literature also suggests that this globalisation notion was easier to adapt as there is a lack of path dependence as to large cities, also called the "exopolis" of suburbs, representing the significant role of exogenous forces in reshaping the city-space (Soja, 2000).

Socio-geographically, Lucy and Philips (1997) view post-suburbanisation as an era in which many suburbs lose their population, their income has decreased relative to the regional situation, and rural, exurban areas far from the developed suburbs are experiencing rapid growth. Another characteristic is the growing suburban area as an economic and social unit such as shopping centres, industrial complexes, office complexes, hospitals, schools and various types of housing scattered (Fishman, 1987). Furthermore, the residents work within or near the suburbs, and the industry gets workers in various service fields.

Additionally, the maturing suburban territory and sprawl settlement of the exurban rural territory also marks this era. Maturity means that housing has been ageing, reducing its attractiveness to middle- and upper-income persons. Selective revival, or gentrification, of some central city neighbourhoods will occur, but inner suburban revival will be unusual once the decline is well advanced because of insufficient neighbourhood non-profit institutions and unesthetic, inconvenient, and unwalkable community designs. As such, planning and design have a critical role in adapting physical characteristics to strengthen territories into places and communities processing this maturity.

We can see the post-suburbanisation with the space fuzzy where the suburb can no longer be seen as a transitional zone between the old city and the open countryside. Rather, it is a new spatial form that is just as lively, attractive, and dynamic as the city (Hudalah & Firman, 2012). Particularly, the identified area is structurally fragmented, economically specialised, and socially segregated (Borsdorf, 2004), which impacts the movement of people that are no longer concentric or radial patterns, but the spatial patterns tend to be polycentric with new suburbs. This shift is also found in social relations identified with the less visible signs of government functionality (Kling, Olin, & Poster, 1991). The post suburban also linked with an "edge city" represented individualism and freedom with automobiles and luxurious private space (Garreau, 1991).

Furthermore, in post-suburbia, the suburbs have loosened their ties to the "central" city. As early as mid 1990s that Savitch (1995) questioned the dependency of suburbs as proven with the increasing

population and income in peripherals compared to the core metropolitan. More recently we have seen growing independent peripheral cities with a variety of specialised activities in addition to residential areas, such as shopping centres, high-tech industrial centres, and educational facilities, eroding the core city's centrality (Fishman, 2002; Soja, 2000). As a result, a multicentred suburban zone emerges, blurring the line between urban and suburban regions (Feng, Zhou, & Wu, 2008). Literature has pointed out how governments have facilitated these growth patterns, such as in the US (Cochrane, 2011; Feng et al., 2008; Phelps & Wood, 2011), in Canada and France with decentralisation (Charmes & Keil, 2015), and in China with a shift of power in the decision-making process of land development (Wu & Phelps, 2011).

The notion also has been confirmed in other countries such as France (Le Goix, 2017) and China (Wu & Phelps, 2008). In France, several issues with implementing post-sub-urbanisation include different criteria in defining urban spaces for study and settlement patterns differences where places of employment have decentralised more slowly in the country. There is also a variation of local governance, as in France, the municipalities have more control over homeowners and the population influx. While in China, the construction of suburban villas and large-scale government affordable housing schemes and suburban industrial development has led the post-sub-urbanisation since the 1990s. Furthermore, this increased population density in the metropolitan area has led to the development of suburban shopping malls and retail parks (Wu & Phelps, 2008).

3. Urban and industrial development in the Jakarta metropolitan area

In this study, to examine post-suburbanisation, we follow the spatial delineation of previous studies by looking at commuters' socio-economy in the Jakarta megacity that includes Special Capital region of Jakarta, with five cities, five districts in the West Java province, and three districts in the Banten province (Aritenang, 2021; Henderson, Kuncoro, & Nasution, 1996; Hudalah & Aritenang, 2017; Hudalah, Viantari, Firman, & Woltjer, 2013) (Fig. 1).

These studies also highlight the abundance of infrastructure that connect these districts, allowing many daily commuters and crucially supporting socio-economic activities in the megacity that included highway, toll road, and commuter train (Fig. 2). The map displays that highways and toll roads stretch from Jakarta to the peripheral areas. However, commuter train lines are only found in the East and South areas, with limited access in the western areas. From the map, we can also distinguish districts in JMA into three categories; the core area that included all Jakarta cities (green), bordering area that included districts that share borders with the core area which are Bekasi regency and city,

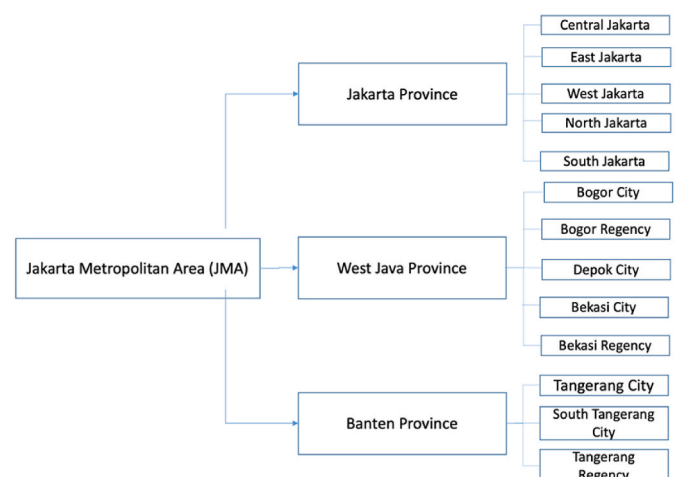


Fig. 1. Districts in JMA.

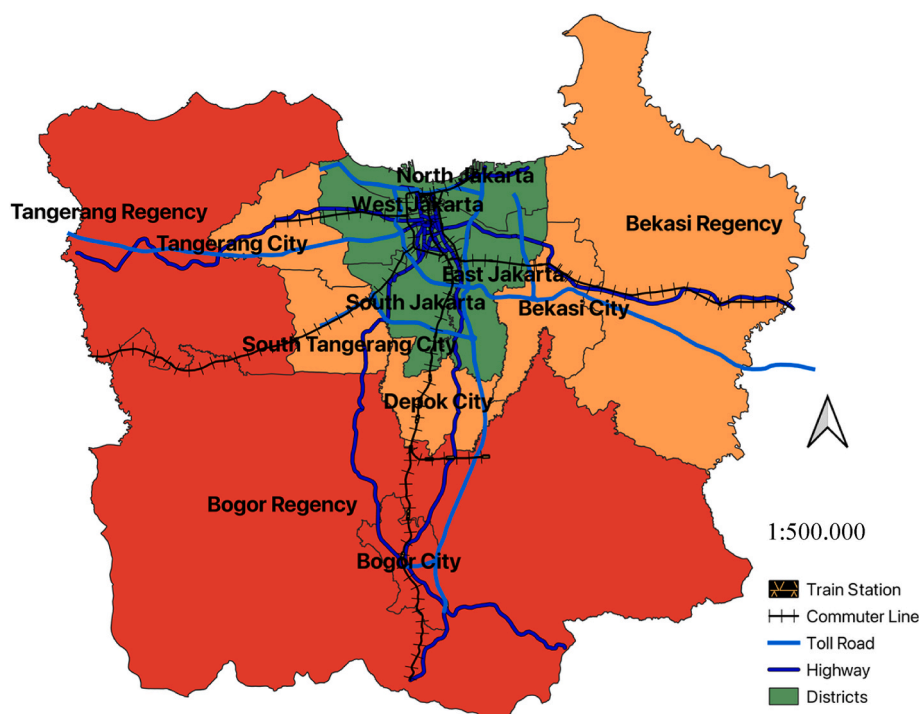


Fig. 2. Districts and infrastructure.

Depok city, South Tangerang and Tangerang cities (orange). Finally, non-bordering districts consist of Bogor city, regency, and Tangerang regency (red).

The role and function of Greater Jakarta as the centre of growth in Indonesia has become more pronounced and significant since the decentralisation from 153 billion Indonesian rupiah (IDR) to 197 billion IDR in 2011 and 2014, respectively. This phenomenon relates to two factors: population growth and economic policy. First, Greater Jakarta continues to develop and urbanise following increasing population and residential area. The population growth followed by housing and economic activities in the region has expanded the build-up area and urbanisation in the region. The result is the specialisation and concentration of economic activities within each district in the region, such as districts bordering Jakarta province providing settlements and commercial areas, using Gross Regional Domestic Product (GRDP) (see Table 1). Second, the economic policy directs the region as the location for manufacturing zones. Presently, further economic activities include warehouses, housing settlements, and industry-linked universities that led to the rapid development that magnifies the region's industry activities and value chain.

The graph above presents the regional inequality by Gini Coefficient of GRDP between districts and cities in metropolitan Jakarta (Fig. 3).

The economic structure of each district shows the specialisation of economic activity in the Jakarta metropolitan area. If we observe the economic manufacturing industry gaps based on value-added, there is a different picture, as we can see a higher inequality level is found in the provinces of Jakarta and West Java. The imbalance occurs in Jakarta due to significant differences between manufacturing industries in East Jakarta and North Jakarta, two administrative cities that have the largest industry in the province. Meanwhile, in the provinces of West Java, Bekasi and Bogor Regencies have higher value-added industries than other districts/municipalities within the province. The last bar suggests an increasing level of regional disparity from 0.445 to 0.446 in 2008 and 2013, respectively. The analysis shows a continued economic disparity growth between districts in the decentralisation period.

The following analysis suggests the specialisation of districts within the megacity. The following graph shows the example of 3 districts, Tangerang City has the highest concentration of leisure and transportation industries, whilst the City Centre of Jakarta hosts the largest concentration of services and construction industry (Fig. 4).

Two factors have contributed to JMA's growth as a metropolitan centre. First, since the October package in the 1990s, the growth of industrial parks, specifically the relocation of industries from Jakarta to suburban areas. Previous studies show that the various industrial

Table 1
GRDP Growth IDR, constant price 2010 (2010–2013) by Districts.

No.	Districts	GRDP (Indonesia Rupiah/IDR)			
		2011	2012	2013	2014
1	Jakarta	1,147,558,226.00	1,222,527,925.00	1,297,195,426.00	1,374,348,612.00
2	Bogor Regency	93,378,723.00	104,286,981.00	110,706,885.00	117,412,136.00
3	Bekasi Regency	51,789,567.09	175,279,801.74	186,479,889.46	197,018,595.08
4	Kota Bogor City	20,766,181.80	23,254,869.50	26,057,306.70	29,102,228.90
5	Kota Bekasi City	43,946,084.12	46,907,332.89	49,739,925.83	52,532,659.96
6	Kota Depok City	29,594,664.90	33,283,562.40	38,517,096.70	43,675,166.90
7	Tangerang Regency	65,741,259.00	72,303,651.00	81,684,833.00	93,312,022.00
8	Tangerang City	31,417,620.00	33,433,900.00	35,411,360.00	37,538,500.00
9	South Tangerang City	33,214,820.00	36,091,800.00	39,290,710.00	4,282,377.00
Total Rata-rata		153,916,604.79	177,115,275.08	189,151,131.75	197,790,530.70

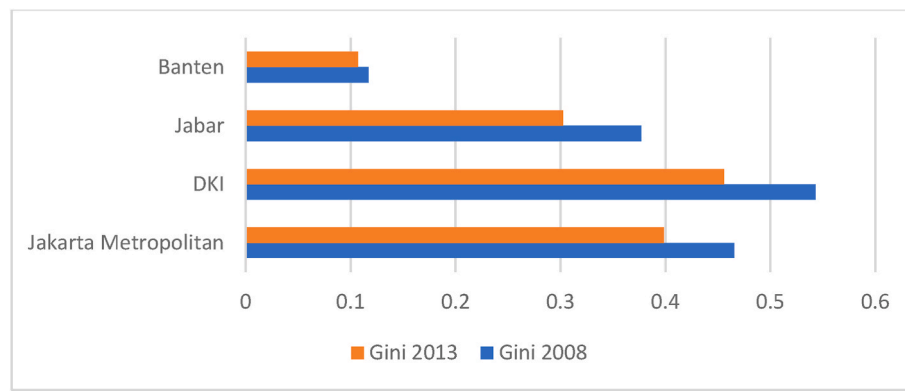


Fig. 3. GRDP Growth, constant price 2010 (2010–2013) by Provinces and Megacity.



Fig. 4. Lq selected districts.

complexes, both large industries (Hudalah & Firman, 2012) and IKM (Hudalah & Aritenang, 2017), can be considered evidence of industrial progress.

Second, the development of population and employment in suburban areas encourages urban activities in various new satellite cities around Jakarta. Periurban development areas with various characteristics such as mixed land use (Rustiadi & Panuju, 2002), residential activities (Winarso, Hudalah, & Firman, 2015) and industrialisation (Firman, 2009, 2011; Firman & Fahmi, 2017; Hudalah & Aritenang, 2017; Hudalah et al., 2013; Hudalah & Firman, 2012). Suburbia in JMA is a result of development opportunities, not spatial planning. Private sector-led new town projects proliferated in the early 1980s, with 23 projects occupying 500–6,000ha each underway or proposed in 1995. Of these, ten were in Tangerang to the west, for example, Serpong (6,000 ha) and Tigaraksa (3,100 ha); four in Bekasi to the east, for example, Cikarang Baru (5,400 ha) and Bekasi (2,000 ha); one in Jakarta and eight in Bogor (Firman, 1998). They are aimed at middle-upper income groups, are often located adjacent to industrial estates, and contain hotels, offices and leisure facilities (e.g. golf courses) (Goldblum & Wong, 2000, p. 32).

Currently, there is evidence of post-suburbanisation with the industrial de-concentration, from the JMA core area to the sub-urban area as a transformation process driven by economic expansion in the form of industrial complexes and new satellite cities (Hudalah, 2019; Rustiadi & Panuju, 2002). There are several empirical studies on suburban issues.

First, suburban spatial transformation in JMA could be regarded as a spatial implication following the restructuring of the agricultural economy towards the manufacturing sector related to global industrialisation and international investment. The core metropolitan area of Jakarta, which is the centre of growth, has reached saturation point, and it is not easy to compete to attract FDI. The emergence of post-suburbia is not entirely the result of government design and cannot fully improvise well (Hudalah & Firman, 2012).

In addition, the case in JMA does not resemble empirical studies in Western countries because many people decide to live in the city centre and commute to suburban areas for work or other activities. This pattern is closely related to the increasing role of the private sector due to government policies that allow the privatisation of land development (Firman & Fahmi, 2017). Second, in the environmental and ecological context, flood problems over the past decade indicate that many areas around the new sub-centre were formed due to post-suburbanisation. This situation occurs significantly in areas with population density growth and significant changes in built-up land areas (Sadewo et al., 2018).

In JMA, industrial and economic development with investment in industrial, trade, and residential development in suburban areas forms the post-suburbanisation. Mainly, the private sector's significant role in the land conversion is clearly with the emergence of gated communities and the construction of new cities on the outskirts of metropolitan Jakarta (Hudalah, Winarso, & Woltjer, 2007; Winarso & Firman, 2002).

that led to spatial and occupational segregation (Winarso et al., 2015). Post-suburbanisation in Indonesia suggests increasing industrial, trade, and residential development (Firman, 2009, 2011). In a comparative study using parametric models to analyse the numbers and location of urban polycentricity for employment density, Sadewo et al. (2021) argue that only the JMA has a polycentric spatial structure.

There are broad studies that examine post-suburbanisation in JMA, such as analysis impact of privatisation of land development (Firman, 1998; Firman & Fahmi, 2017; Hudalah et al., 2013; Winarso et al., 2015), metropolitan governance and policy (Hudalah, 2017; Rukmana, 2015; Sadewo, Syabri, & Pradono, 2020) and the geographical development of the region (Hudalah & Firman, 2012; Rustiadi et al., 2021) encourage post-trade development. As such, the above studies argue post-suburbanisation in JMA shows mixed patterns characteristic of traditional and new housing development due to central government policies that attract foreign investment flows (FDI).

However, studies show that the new towns segregate middle-upper people from most of the metropolitan region's population. There is also segregation between middle-upper income areas within the towns, produced by differences in quality, price and level of security and reinforced by internal controls on movement between neighbourhoods (Firman, 1998). For instance, the development of the Bekasi new town and the adjacent industrial estate led to the removal of 16,500 residents and 12,000 farmers (Goldblum & Wong, 2000), while during the development of Tigaraksa, about 1,400 land-owning agricultural households were evicted (Firman, 1998). These new towns managed by the private developers use American-influenced design concepts; aim at high quality, often luxury, houses and surroundings; often seek to be self-sufficient for utilities, and attract prosperous families searching for security and improved lifestyles.

4. The post-suburbanisation transformation in Jakarta metropolitan area

This paper uses the commuting survey datasets organised by the Central Agency on Statistics using a questionnaire as weighted data of 12,960 and 13,120 daily commuters representing 2.9 million and 3.2 million individuals above 15 years old living in the JMA in 2014 and 2019, respectively. The JMA commuting surveys include thirteen districts which are eight municipalities and three regencies from three different provinces within the (JMA); DKI Jakarta Province (Central Jakarta, North Jakarta, South Jakarta, East Jakarta, and West Jakarta), West Java Province (Depok, Bogor, Bogor City, Bekasi, and Bekasi City), and Banten Province (i.e. Tangerang, Tangerang City, and South Tangerang).

The survey uses a spatial stratified sampling strategy, using the two-stages sampling method at the subdistrict level by considering household and population size distribution based on Indonesia's 2010 census data (Sofiyandi & Siregar, 2020). The survey has comprehensive information that allows the study to examine post-suburbanisation transformation using commuter and their behaviour related to post-suburbanisation aspects such as origin-destination districts, activities of commuting such as work, education and shopping, time and hour of commuting, and commuters' socio-economic characteristics. This study examines post-suburbanisation process by examining commuters' socio-economy in several factors; spatial independence, maturity, and segregation using commuting patterns and socio-economic dynamics descriptive analysis.

Based on their economic structure, the districts can be classified into three groups. First, the non-autonomous Jakarta cities as the core of the JMA are Central, West, East, South and North Jakarta. These core areas are surrounded by five districts that share borders and serve as the hosts' industrial zones: Depok City, Bekasi City, Bekasi Regency, Tangerang city, and South Tangerang City. The other districts tend to be dominated by residential and light industrial zones, with mixed land use and agricultural activities.

4.1. Spatial independence

First, we analyse the independence of peripheral to core areas by examining the share of commuters from each district in the core and peripheral areas. The commuting network is presented in the following table, suggesting the much higher share of commuters heading to the core areas between peripheral residents compared to the core residents suggesting the intense commuting activity in the JMA (Table 2).

The table suggests that South and Central Jakarta is the leading destination with more than 50 per cent of commuters entering these cities and has been identified as the centre of financial and business services since the mid-1990s (Henderson et al., 1996). These cities host main offices such as *Thamrin*, *Sudirman* and *Sudirman Central Business District*. This role also applies to both other parts of core areas and peripherals. For instance, commuters from East and West Jakarta, known for its numerous public housing cities developed in the 1980s and 1990s, head for South and Central Jakarta. Similarly, districts from peripherals such as Depok city, Bogor regency, and South Tangerang cities have more than 43 per cent of their daily activities in South Jakarta. Whereas more than 32 per cent of commuters from Bogor regency and Bogor city heads to Central Jakarta. South and Central Jakarta are known as the centre for leisure and business zones such as *Kemang* and *Panglima Polim* (sub-districts that are known for youthful restaurants and cafés, and residents of high-class foreigners), Menteng (sub-district that hosts cultural, historical landmarks, restaurants, and residents of high-ranking officials), and *Sudirman-Thamrin* is the main road that serves as a business centre with the vibrant night life of bars and clubs (Tadié & Permana-deli, 2015).

The data shows that more than 40.2% of commuters heading to the core area are from the peripheral area. Most commuters originated from Depok and Bekasi cities, more than 43.8%, with a total of 588,639 commuters. The figures also reveal that commuters from the peripheral areas head to the closest part of the core area, for instance, more than 61.6% of both Bekasi districts commuters commute daily to East Jakarta, whilst most commuters from Depok head to South Jakarta (53.6%). Similarly, more than 60% of commuters to West Jakarta came from Tangerang regency and cities. This finding indicates the dependence of bordering districts on core areas.

On the other hand, there are only 7.7% of daily commuters originating from the core area to peripherals. The largest number of commuters are from East and South Jakarta, who travel to their nearest peripheral districts, Bekasi and South Tangerang cities, respectively. While there are 26.7% of commuters travelling between peripherals suggesting the presence of post-suburbanisation, confirming a variety of specialised activities in the peripherals and emerging multicentred suburban zones (Fishman, 2002; Soja, 2000; Feng, Zhou, & Wu, 2008).

If we examine the commuting purpose, bordering districts have

Table 2
Number of commuters, origins and destinations by Districts, 2019

Districts	Core	Peripherals	Total
Core			
South Jakarta	154,251	76,725	230,976
East Jakarta	254,299	89,885	344,184
Central Jakarta	93,507	8,861	102,368
West Jakarta	219,849	65,417	285,266
North Jakarta	128,172	13,960	142,132
Peripherals			
Bogor Regency	204,741	278,312	483,053
Bekasi Regency	115,136	110,243	225,379
Bogor City	21,514	64,833	86,347
Bekasi City	282,086	92,102	374,188
Depok city	306,553	99,392	405,945
Tangerang Reg	78,206	149,731	227,937
Tangerang City	175,064	61,294	236,358
South Tangerang City	160,143	37,259	197,402
Total	2,193,521	1,148,014	3,341,535

higher dependence, both work and school, on the core area (Table 3). The figure suggests that as high as 25 per cent and 20 per cent of commuters from Bekasi and Depok cities work in Jakarta, respectively. Similarly, more than 23 per cent of students from both cities go to school in Jakarta. Despite not being as high as its neighbours from West Java province, bordering districts from Banten province, Tangerang and South Tangerang cities, more than 13 per cent of commuters from both cities work and study in Jakarta. On the other hand, less than 13 per cent and 8 per cent of commuters from non-bordering districts; Bogor regency, Bogor city and Tangerang regency work and study in Jakarta, respectively.

4.2. Maturity

Second, the maturity of settlement with ageing housing, loss of attractiveness to middle- and upper-income families, and gentrification signs reflect post-suburbs transformation. First, we examine the average age of commuters from the peripherals. Commuters tend to be older in non-bordering districts such as Bogor city (39.5 years old) and Tangerang regency (40.4) compared to bordering districts; Depok city (33.2), and Bekasi city (34.5). Furthermore, younger people are found in districts in the Banten province, such as Tangerang and South Tangerang cities, 34.7 and 35.8 years old, respectively, reflecting the more recent real estate development in these districts. If we cross-analyse with the previous table, districts with younger average commuters would have a higher share of commuters towards the core areas such as Bekasi, Depok, Tangerang and South Tangerang cities. On the contrary, districts with older average commuters have a lower share of commuters to Jakarta, such as the Bekasi regency, Bogor city and Tangerang regency. This finding may indicate age is an important determinant for commuting with young people tending to commute longer and further, as found elsewhere (Eurostat, 2016; Newbold, 2022, pp. 1–16).

Next, we analyse the maturity of suburbs using two approaches; the attractiveness of suburbs and the presence of gentrification. First, we examine the maturity of suburbs by looking at the share of new migrants and middle-upper-class (Table 4). The literature explains that ageing properties and loss of attractiveness for the middle-upper-class indicate suburb maturity. The data on new arrivals within the five years shows that Depok City, Bogor Regency and Tangerang City have the highest number of migrants, respectively. With more than 13 thousand new commuters, these districts were also the highest proportion of migrants in the same period. Second, gentrification following the middle-upper class preferences is found in Bogor regency and Depok city with more than 12.2 and 7.8 per cent of commuters in the districts, respectively. Together with Tangerang city, these bordering districts are relatively dynamic, with younger properties following new migrants. Only Bogor city has a considerable share of the middle-upper class at 8.29%; however, the number of new migrants is very low, only 701 commuters. The table also suggests peripherals are more attractive for middle-upper class migrants, with the average share of middle-upper migrants being higher than the overall average share of migrants, 4.8 and 5.4 per cent, respectively.

Compared with the infrastructure availability, the presence of

highways attracted more new migrants to the peripheral areas except for South Tangerang city, as found in Yudhistira, Indriyani, Pratama, Sofiyandi, and Kurniawan (2019). The South Tangerang city, without a highway directly from Jakarta but connected with a toll road and commuter train line, had relatively low migrants from within JMA, suggesting the highway remains preferable by commuters in JMA. Furthermore, local government policy may determine the migration destination. For instance, the table suggests migration to West Jakarta was significantly high, especially the number and share of middle-upper class, compared with other parts of Jakarta. This finding could be explained by the province's Strategic Development Plan to promote the east-west development orientation with mega shopping malls, such as Taman Anggrek Mall and Puri Indah Mall, that expands toward the west (Han & Basuki, 2001). Consequently, this led to a new urban image and increased land values, attracting more middle-upper class to relocate to West Jakarta.

Third, we examine the presence of gentrification using the proportion of young and middle-upper-class commuters (Table 5). The share of young commuters was found the most in East Jakarta, compared to other districts within the JMA, which sustains in 2019. While in the peripherals, Depok, Tangerang, and South Tangerang cities have the highest share of young commuters for both 2014 and 2019. On the other hand, the non-bordering districts, Bogor city and Tangerang regency, continuously had the least share of young commuters.

However, a different picture is found in the share of middle-upper-class commuters, whereas these non-bordering districts had the highest share of middle-upper-class commuters in the JMA, with more than 42 and 39 per cent, relatively. Furthermore, Tangerang and South Tangerang cities had the lowest share of middle-upper-class commuters among the peripheral areas. These continuous contradictions between the share of young and middle-upper-class commuters suggest gentrification is found in different districts. Bordering districts are more attractive for young commuters, and non-bordering districts are more attractive for middle-upper-class commuters.

On the other hand, the table also suggests the low share of middle-upper-class commuters in the core area, especially in Central and West Jakarta, with less than 17 per cent of commuters. This finding suggests the maturity of peripherals areas as the share of young and middle-upper class are higher than the core areas, indicating peripherals are more attractive as an alternative residential area for the JMA residents.

4.3. Spatial segregation

The third indicator is the presence of socio-economic segregation in JMA. The following table shows the average wage, car share and formal worker share of commuters toward the core JMA (Table 6). Income gaps are obvious between bordering and non-bordering districts. In 2014, the bordering districts had relatively higher average wages with less than 5 million IDR, whilst the non-bordering district's wages were more than 7 million IDR. The figures imply that the residential and urban development in JMA has grown at the expense of regional inclusiveness. The data also reveals the significant spatial segregation where Bogor city had the highest average wage and Bekasi city has the highest share of decent housing.

Another spatial segregation is the number and size of new towns by the developers may influence the poverty rates among districts in the Jakarta metropolitan. For instance, districts with a higher share of formal workers, such as Depok city, and South Tangerang city and Tangerang city have a lower poverty rate, 2.32%, 1.68% and 4.91%, respectively (BPS, 2016). This finding confirms a previous study by Rukmana and Ramadhani (2021) that reveals a concentration increase of the top occupation group (TOG) in Depok and South Tangerang City following the development of luxury apartments such as De Vonte Apartments and Permata Green Sentosa (Depok City) and new central business district, and middle-upper class housing, especially in Bumi Serpong Damai (BSD) (South Tangerang city).

Table 3
Share of commuters by Activities, Commuters to core JMA 2014 and 2019 (%).

	Work		School	
	2014	2019	2014	2019
Peripheral				
Bogor Regency	11.28	12.73	3.26	7.70
Bekasi Regency	8.69	9.35	7.35	7.69
Bogor City	2.40	1.62	0.13	0.59
Bekasi City	24.87	21.72	31.76	24.03
Depok city	19.85	21.73	23.88	32.48
Tangerang Reg	4.15	6.38	2.21	1.53
Tangerang City	14.07	13.33	13.31	15.54
South Tangerang City	14.69	13.14	18.11	10.44

Table 4

Destinations and Proportion of Migration, Commuters to core JMA 2019.

	Destinations of Migrant within JMA in 5 years (people)		Number of Commuters in 2019 (people)		Proportion of migrants within 5 years (%)	
	All	Middle-Upper	All	Middle-Upper	All	Middle-Upper
South Jakarta	10,027	1,148	154,251	38,857	6.50	2.95
East Jakarta	7,011	1,594	254,299	74,824	2.76	2.13
Central Jakarta	6,828	619	93,507	15,662	7.30	3.95
West Jakarta	9,766	4,066	219,849	36,064	4.44	11.27
North Jakarta	7,333	2,616	128,172	26,637	5.72	9.82
Bogor Regency	14,468	5,931	204,741	48,533	7.07	12.22
Bekasi Regency	6,955	1,505	115,136	37,373	6.04	4.03
Bogor City	701	701	21,514	8,461	3.26	8.29
Bekasi City	8,894	1,628	282,086	68,328	3.15	2.38
Depok city	19,961	6,301	306,553	80,739	6.51	7.80
Tangerang Reg	4,097	1,088	78,206	36,192	5.24	3.01
Tangerang City	13,559	2,292	175,064	37,813	7.75	6.06
South Tangerang City	0	0	160,143	35,293	0.00	0.00
Total	109,600	29,489	2,193,521	544,776	5	5.69

Table 5

Proportion of Young and middle-upper-class commuters, Commuters to core JMA 2014 and 2019 (%).

	Proportion of Resident in 2014		Proportion of Resident in 2019	
	Young	Middle-Upper	Young	Middle-Upper
South Jakarta	75.62	25.85	73.76	25.19
East Jakarta	81.42	29.54	81.75	29.42
Central Jakarta	75.80	18.86	74.61	16.75
West Jakarta	76.14	16.41	77.02	16.40
North Jakarta	77.04	20.47	76.49	20.78
Bogor Regency	47.88	22.35	50.36	23.70
Bekasi Regency	68.19	32.84	70.97	32.46
Bogor City	50.35	42.03	49.72	39.33
Bekasi City	68.15	23.93	68.85	24.22
Depok city	71.63	26.53	72.04	26.34
Tangerang Reg	49.16	47.90	51.34	46.28
Tangerang City	70.49	21.68	70.94	21.60
South Tangerang City	69.93	21.82	70.06	22.04

Table 6

Socio-Economic Segregation, Commuters to core JMA 2014 and 2019.

	Wage (IDR)		Decent Housing Proportion (%)		Formal Worker Proportion (%)	
	2014	2019	2014	2019	2104	2019
Peripheral						
Bogor Regency	5005781	5127979	70.16	68.18	96.53	100.00
Bekasi Regency	3903742	5488587	82.28	88.39	97.28	95.79
Bogor City	4516667	8398174	73.68	76.27	97.33	93.62
Bekasi City	5898324	6031143	82.32	83.93	97.78	99.65
Depok city	4430003	6162676	77.90	70.55	98.40	98.35
Tangerang Reg	2911719	6044033	73.97	81.82	92.19	96.92
Tangerang City	5061748	5140476	75.50	74.53	96.84	99.06
South Tangerang City	5368843	6720784	80.05	72.73	98.22	99.08

To develop these new towns, developers have to prepare and convince the government to change the land use. Thus, developers are required to consult the right person and government agency for support on infrastructure development. Previous studies have identified this with concerns over security issues among members of the upper-and middle-class society that live in gated communities around the industrial estates, such as in the new towns of Jababeka, Lippo and Deltamas, located in Bekasi regency (Hudalah & Firman, 2012). The study also suggests a lack of communication between the industrial estate companies that focus on accommodating groups of investors from a particular country of origin, such as Japanese (MM2100, BFIE, and EJIP) and Korean (BIIE). Furthermore, these industrial zones led to increasing spatial inequalities between urban areas with specific urban and

industrial infrastructure, such as road networks, telecommunication networks, wastewater treatment plants, and clean water treatment plants (Hudalah, 2017; Hudalah, Nurrahma, Sofhani, & Salim, 2019).

Transportation modes of commuters also represent segregation, whereas low-income people prefer to spend less than 10 per cent of their weekly incomes on transport (Carruthers, Dick, & Saukar, 2005). Thus, understanding the proportion of commuter transportation modes is critical to examine the post-suburbanisation transformation by identifying the government's role in transportation infrastructure. In 2019, the table below suggests that private vehicles (motorcycles and cars) dominated the transportation modes with more than 60% and 8.8%, respectively (Table 7). Commuters also use trains that operate from Bogor, Bekasi and Depok cities. The hail riding motorcycle and car also plays a critical role, with more than 4.9% of commuters from the peripheral using these modes of transportation to the core.

The figure also suggests that more commuters use ride-hailing, and TransJakarta is from bordering districts such as Bekasi, Depok, and Tangerang cities. The figures suggest the importance of public infrastructure in post-suburbanisation with the increasing number of commuters using Train and TransJakarta. At the same time, the growth of motorcycles has reduced, and the number of car users has dropped significantly. Interestingly, for outer peripherals, such as Bogor and Tangerang regencies, the number of commuters within peripherals are much higher than the core area, suggesting the post-suburbanisation process in the JMA.

Our findings suggest clear socio-economic segregation regarding income, decent housing, and transportation modes in the JMA. There are higher wages and a higher share of decent housing in bordering districts, suggesting relatively lower-income commuters reside in non-bordering districts. Furthermore, bordering districts enjoy less cost and more alternative transportation means such as motorcycle, car and TransJakarta due to spatial proximity. As commuters from non-bordering districts are lower-income, they rely more on public transportation such as buses and trains.

5. Discussion and conclusion

This study reveals that the post-suburbanisation process in the JMA occurs in commuting scales and patterns. The study findings suggest the process of post-suburbanisation where suburbans that were peripherals transforms as new centres as part of the ongoing polycentric development process, as indicated elsewhere (Firman & Fahmi, 2017; Hudalah & Firman, 2012; Phelps, 2018; Phelps, 2018, 2018).

First, the metropolitan core is experiencing low population growth, and peripheral areas are growing in both population and economic growth. Following the high living costs and land prices, our study shows Jakarta City, as a core region, remains the main city for economic and social activities. Meanwhile, suburban is converted to a land-use mix

Table 7

Share of Transportation Mode, from periphery to core JMA 2014 and 2019 (%).

	Non motorized	Motor	Hail Riding	Car	Hail Riding-car	Public Transportation	Train	TransJakarta
Bogor Regency	0.0	8.7	4.2	9.3	10.2	16.7	21.2	28.5
Bekasi Regency	0.0	8.0	4.5	5.8	9.0	10.7	14.2	11.8
Bogor City	0.0	0.2	0.0	4.1	1.4	0.4	5.2	0.0
Bekasi City	0.0	22.8	29.3	26.5	30.9	26.9	14.6	15.6
Depok city	53.1	23.6	17.6	27.6	15.5	22.6	24.2	16.3
Tangerang Reg	0.0	5.4	3.8	5.1	19.6	12.8	2.9	2.9
Tangerang City	0.0	17.6	23.2	6.6	7.2	7.2	5.4	15.8
South Tangerang City	46.9	13.6	17.4	15.0	6.2	2.7	12.3	9.0
Average	0.1	60.0	2.7	8.8	2.2	5.8	16.9	3.4

with new towns and large-scale residential areas and agricultural land. As such, this study confirms how these once dormitory towns have transformed into large cities with all urban activities, except working places.

Second, the study found that the declining commuting time may suggest post-suburbanisation with the expansion of urban polycentric and the declining attractiveness of non-bordering suburban areas. Spatial proximity allows lower commuting costs and more alternative transportation modes. Whilst districts in West Java are more connected with the core areas as seen with higher activities and availability of public transportations such as the commuter train.

Our study reveals that 40.2% of commuters from the peripherals are heading to the core area, slightly above the 35% threshold of dominantly monocentric, as [Bertaud \(2009\)](#) suggested. The result may imply the growing independence of suburban districts following the presence of new peripheral centres. However, the dependence of these suburban districts with its neighbouring core city remains high, such as almost 62% per cent of both Bekasi districts commuters travel daily to East Jakarta and more than 53% of Depok city commuters head to South Jakarta (53.6%). On the other hand, there are 893,166 commuters (26, 7%) that travel within peripheral areas, which implies a considerable size of commuters that are not dependent on economic activities in the core. The highest share of commuters from peripheral districts that travel for work are from Bekasi and Tangerang cities (about 90%), whilst for schools are from Bekasi regency (40%).

Third, the study findings also indicate the gentrification and segregation in the JMA following neoliberalism development of urban revitalisation and new development by the private sector ([Herlambang, Leitner, Tjung, Sheppard, & Anguelov, 2019](#); [Sheppard, 2018](#); [Winarso et al., 2015](#)) and unbalanced infrastructure provision by the government. The study identified selective revival and gentrification, especially in border districts, due to spatial proximity to the core. The study reveals that different types of gentrification occur in different locations. While the gentrification of younger people is in bordering districts, the gentrification of the middle-upper class is found in non-bordering districts, such as Bogor city and Tangerang regency, with commuters' having higher wages and more decent housing. Furthermore, South Tangerang is an interesting case; it had the highest average wage and the highest share of decent housing. The districts are favourable for first-time middle-upper class homeowners, as suggested with a high share of young people and a relatively high number of cars, confirming the city development is shaped by foreign and private investors, as argued by [Shatkin \(2019\)](#).

Finally, our study confirms the importance of infrastructure provision for public transportation on population growth, as suggested by [Chirfield and Panggabean \(1995\)](#). The lower share of commuters from Banten districts (Tangerang and South Tangerang cities) to the JMA core indicates a lack of train and other public transportation. More than 75% of commuters use private vehicles. On the other hand, less than 70% of commuters from West Java districts (Bogor, Depok and Bekasi cities) use private vehicles. Our findings indicate that a higher share of commuters resides in bordering districts, confirming [Firman and Fahmi \(2017\)](#) found a mixed pattern of traditional and new suburban residential

development in the JMA post-suburbanisation process. It is unlikely to fully resemble Western cities, perhaps due to many people in the JMA choosing to live in the traditional core and commuting out to suburban developments for activities, suggesting the persistent link with the main cities.

The paper contributes to the post-suburbanisation debate and policy by highlighting the importance of multilevel governance in managing megacities' variation of commuting mobility, spatial and socio-economic development. First, our study provides evidence that post-suburbanisation could be identified through the utilisation of commuting data as innovative and alternative data. The commuting pattern and commuters' socio-economic characteristics suggest that suburbs have detached from core cities and transformed as new centres following each peripherals-respective specialised development.

Second the paper highlights the importance of strengthening inter-government institutional capacity to formulate, coordinate, and implement better industrial development and residential regulations along major inter-city highways. The metropolitan governance is critical to ensure the integrated development of industries, housing, jobs, shopping, educational opportunities, and infrastructure. The paper shows that polycentric development reduces commuting time and congestion which requires better urban governance capacities. The process of gentrification, revitalisation, new urban development and connectivity between these new urban centres should be managed more effectively in cross-government districts. The establishment of Greater Jakarta Transportation Agency (*Badan Pengelola Transportasi Jabodetabek/BPTJ*) by the Transportation Ministry allows to manage the JMA transportation management, such as the operation of TransJakarta feeder bus in Jakarta districts and Residence Connexion (JR Connexion) bus in Bogor districts.

Third, our study suggests the importance of the government's role in ensuring inclusive and sustainable development in the JMA. Presently, the private sector, with its gentrification and new town development across the JMA, has led to post-suburbanisation. The lack of effective spatial planning and public transportation potentially deepens large real-estates and gated communities and, thus, segmented peripheral areas.

Author statement

Adiwan Aritenang: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Validation; Visualization; Roles/ Writing - original draft; Writing - review & editing.

Declaration of competing interest

None.

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